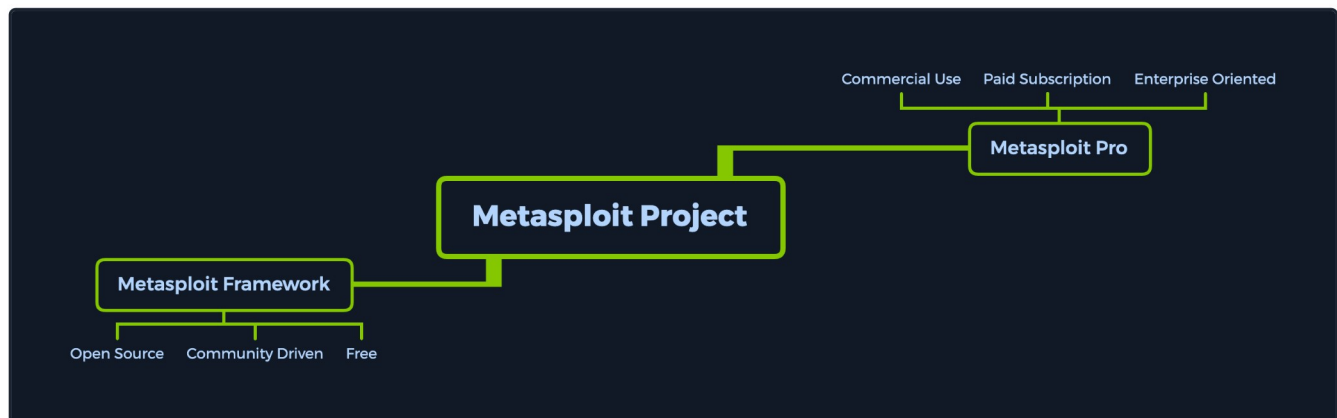


Introduction to Metasploit

The **Metasploit Project** is a Ruby-based, modular penetration testing platform that enables you to write, test, and execute the exploit code. This exploit code can be custom-made by the user or taken from a database containing the latest already discovered and modularized exploits. The **Metasploit Framework** includes a suite of tools that you can use to test security vulnerabilities, enumerate networks, execute attacks, and evade detection. At its core, the **Metasploit Project** is a collection of commonly used tools that provide a complete environment for penetration testing and exploit development.



The **modules** mentioned are actual exploit proof-of-concepts that have already been developed and tested in the wild and integrated within the framework to provide pentesters with ease of access to different attack vectors for different platforms and services. Metasploit is not a jack of all trades but a swiss army knife with just enough tools to get us through the most common unpatched vulnerabilities.

Its strong suit is that it provides a plethora of available targets and versions, all a few commands away from a successful foothold. These, combined with an exploit tailor-made to those vulnerable versions and with a payload that is sent after the exploit, which will give us actual access into the system, provide us with an easy, automated way to switch between target connections during our post-exploitation ventures.

Metasploit Pro

Metasploit as a product is split into two versions. The **Metasploit Pro** version is different from the **Metasploit Framework** one with some additional features:

- Task Chains
- Social Engineering
- Vulnerability Validations
- GUI
- Quick Start Wizards
- Nexpose Integration

If you're more of a command-line user and prefer the extra features, the Pro version also contains its own console, much like **msfconsole**.

To have a general idea of what Metasploit Pro's newest features can achieve, check out the list below:

Infiltrate

Collect Data

Remediate

Manual Exploitation	Import and Scan Data	Bruteforce
Anti-virus Evasion	Discovery Scans	Task Chains
IPS/IDS Evasion	Meta-Modules	Exploitation Workflow
Proxy Pivot	Nexpose Scan Integration	Session Rerun
Post-Exploitation		Task Replay
Session Clean-up		Project Sonar Integration
Credentials Reuse		Session Management
Social Engineering		Credential Management
Payload Generator		Team Collaboration
Quick Pen-testing		Web Interface
VPN Pivoting		Backup and Restore
Vulnerability Validation		Data Export
Phishing Wizard		Evidence Collection
Web App Testing		Reporting
Persistent Sessions		Tagging Data

Metasploit Framework Console

The `msfconsole` is probably the most popular interface to the [Metasploit Framework \(MSF\)](#). It provides an "all-in-one" centralized console and allows you efficient access to virtually all options available in the [MSF](#). `Msfconsole` may seem intimidating at first, but once you learn the syntax of the commands, you will learn to appreciate the power of utilizing this interface.

The features that `msfconsole` generally brings are the following:

- It is the only supported way to access most of the features within [Metasploit](#)
- Provides a console-based interface to the [Framework](#)
- Contains the most features and is the most stable [MSF](#) interface
- Full readline support, tabbing, and command completion
- Execution of external commands in `msfconsole`

Both products mentioned above come with an extensive database of available modules to use in our assessments. These, combined with the use of external commands such as scanners, social engineering toolkits, and payload generators, can turn our setup into a ready-to-strike machine that will allow us to seamlessly control and manipulate different vulnerabilities in the wild with the use of sessions and jobs in the same way we would see tabs on an Internet browser.

The key term here is usability—user experience. The ease with which we can control the console can improve our learning experience. Therefore, let us delve into the specifics.

Understanding the Architecture

To fully operate whatever tool we are using, we must first look under its hood. It is good practice, and it can offer us better insight into what will be going on during our security assessments when that tool comes into play. It is essential not to have [any wildcards that might leave you or your client exposed to data breaches](#).

By default, all the base files related to Metasploit Framework can be found under `/usr/share/metasploit-framework` in our [ParrotOS Security](#) distro.

Data, Documentation, Lib

These are the base files for the Framework. The Data and Lib are the functioning parts of the `msfconsole` interface, while the Documentation folder contains all the technical details about the project.

Modules

The Modules detailed above are split into separate categories in this folder. We will go into detail about these in the next sections. They are contained in the following folders:

Modules
<pre>dbcypth0n@htb[/htb]\$ ls /usr/share/metasploit-framework/modules</pre>
<pre>auxiliary encoders evasion exploits nops payloads post</pre>

Plugins

Plugins offer the pentester more flexibility when using the `msfconsole` since they can easily be manually or automatically loaded as needed to provide extra functionality and automation during our assessment.

Plugins
<pre>dbcypth0n@htb[/htb]\$ ls /usr/share/metasploit-framework/plugins/</pre>
<pre>aggregator.rb ips_filter.rb openvas.rb sounds.rb alias.rb komand.rb pcap_log.rb sqlmap.rb auto_add_route.rb lab.rb request.rb thread.rb beholder.rb libnotify.rb rssfeed.rb token_adduser.rb db_credcollect.rb msfd.rb sample.rb token_hunter.rb db_tracker.rb msgrpc.rb session_notifier.rb wiki.rb event_tester.rb nessus.rb session_tagger.rb wmap.rb ffautoregen.rb nexpose.rb socket_logger.rb</pre>

Scripts

Meterpreter functionality and other useful scripts.

Scripts
<pre>dbcypth0n@htb[/htb]\$ ls /usr/share/metasploit-framework/scripts/</pre>
<pre>meterpreter ps resource shell</pre>

Tools

Command-line utilities that can be called directly from the `msfconsole` menu.

Tools
<pre>dbcypth0n@htb[/htb]\$ ls /usr/share/metasploit-framework/tools/</pre>
<pre>context docs hardware modules payloads</pre>

dev

exploit

memdump

password

recon

Now that we know all of these locations, it will be easy for us to reference them in the future when we decide to import new modules or even create new ones from scratch.

Questions

[Cheat Sheet](#)

Answer the question(s) below to complete this Section and earn cubes!

+ 0

Which version of Metasploit comes equipped with a GUI interface?

Submit your answer here...



Submit

+ 0

What command do you use to interact with the free version of Metasploit?

Submit your answer here...



Submit

← Previous

Next →

[Cheat Sheet](#)

? [Go to Questions](#)

Table of Contents

Introduction

Preface



[Introduction to Metasploit](#)

Introduction to MSFconsole

MSF Components

Modules

Targets

Payloads

Encoders

Databases

Plugins & Mixins

MSF Sessions

Sessions & Jobs

Meterpreter

Additional Features

Writing & Importing Modules

Learning Objectives

Introduction to MSFVenom

Firewall and IDS/IPS evasion

Metasploit-Framework Updates - August 2020

My Workstation

O F F L I N E

▶ Start Instance

∞ / 1 spawns left